# Visited Countries and Cities

**Filter ( *Finland* )** **All** **Finland** **Lebanon** **South Korea**

| COUNTRY | CITY | POPULATION |
|---|---|---|
| Finland | Espoo | 213271 |
| Finland | Helsinki [Helsingfors] | 555474 |
| Finland | Lahti | 96921 |
| Finland | Oulu | 120753 |
| Finland | Tampere | 195468 |
| Finland | Turku [Åbo] | 172561 |
| Finland | Vantaa | 178471 |

**Fig 1:** List of visited cities in Finland

**GET EXAMPLE 1:**

**app.js:**

```
import express from "express";
import {countries, cities} from "./db.js"

const app = express();
app.set("view engine", "ejs");
app.use(express.urlencoded({ extended: true}))


app.get("/", (req, res) => {

    let result = Object.values(cities);
    const filterBy = req.query.filterBy ??  "ALL" ;

    if (filterBy === "ALL")
    {
        res.render("q1", {filterBy, cities, countries, result})
    }
    else{
     result = result.filter(x => x.code ===  countries.find(c => c.name
=== filterBy).code )
     res.render("q1", {filterBy, cities, countries, result})
    }



})


app.listen(3000, () => {
    console.log("server is running");
})
```

**q1.ejs**

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        a {
            color: green;
            font-weight: bold;
            margin: 10px;
            text-decoration: none;
        }

        div {
            text-align: center;
        }

        table {
            border-collapse: collapse;
            margin: 20px auto
        }

        th,
        td {
            border: 1px solid black;
            padding: 8px
        }

        h1,
        p {
            text-align: center;
        }
        span.black {
            color: black;
            font-weight: bold;
        }
    </style>
```

```
</head>

<body>
    <h1>
        VISITED COUNTRIES AND CITIES
    </h1>
    <div class="Countries">

        <span class="black" > Filter(<%=
            filterBy;
            %>)</span>

         <% if (filterBy !== "ALL"){ %>
            <span>
                <a href="/?filterBy=ALL">ALL</a>
            </span>
            <% } %>
        <%
        countries.forEach(c=> {
            if (c.name !== filterBy){
            %>
            <span>
                 <a href="/?filterBy=<%=c.name %>"><%=c.name %> </a>

            </span>
            <%} }); %>

    </div>
    <div>
        <table>
            <thead>
                <tr>
                    <th>COUNTRY</th>
                    <th>CITY</th>
                    <th>POPULATION</th>
                </tr>
            </thead>
            <tbody>
                <tr>
                    <% result.forEach(r=> {
```

```
                        %>
                    <td><%= countries.find(c => c.code === r.code )?.name
%></td>
                    <td><%= r.name %></td>
                    <td><%= r.population%></td>
                         </tr>
                <%  });   %>
            </tbody>
        </table>


    </div>
</body>

</html>
```

Product: S25 Ultra

Units: 2

Storage: 256GB 512GB 1TB

Shipping:

Buy

S25 Ultra x 2 purchased, Total price is **140500**

**Fig 2** : Total price

**POST EXAMPLE 1:**

**q2.js**

```
import express from "express";
import { body, validationResult } from "express-validator";
const app = express();
app.set("view engine", "ejs");

app.use(express.urlencoded({ extended: true }));

app.get("/", (req, res) => {
  res.render("q2", { errors: {}, form: {}, price: null });
});

app.post("/", [
  body("units").isInt({ min: 1 }).withMessage("Error: The Unit should be a
positive integer number"),
  body("storage").exists().withMessage("Error: Select a storage"),
], (req, res) => {
  const errors = validationResult(req);
  let price , model ;
  if (errors.isEmpty()) {
      const basePrice = [70000, 65000, 50000]
      const storage = [1, 1.1, 1.25]
      price = basePrice[req.body.model] * storage[req.body.storage] *
req.body.units;
      price += req.body.shipping ? 500 : 0;
      const models = ["S25 Ultra", "S25 Edge", "S25+"]
      model = models[req.body.model]
    //   console.log(price)
  }
  res.render("q2", { errors: errors.mapped(), form: req.body, price, model
});
});

app.listen(3000, () => console.log("Listening on http://localhost:3000"));
```

**q2.ejs:**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        table { margin: auto;}
        td { padding: 0.5em; }
        p { text-align: center;}
        .error { text-align: center; color: red; font-style: italic;}
        span { color: blue}
    </style>
</head>
<body>
    <form action="/" method="post">
        <table>
            <tr>
                <td>Product:</td>
                <td>
                    <select name="model" id="">
                        <option value="0" <%= form.model == "0" ?
"selected" : "" %> >S25 Ultra</option>
                        <option value="1" <%= form.model == "1" ?
"selected" : "" %> >S25 Edge</option>
                        <option value="2" <%= form.model == "2" ?
"selected" : "" %> >S25+</option>
                    </select>
                </td>
            </tr>

            <tr>
                <td>Units:</td>
                <td>
                    <input type="text" name="units"  value="<%= form.units
?? '' %>">
                </td>
            </tr>
```

```
            <tr>
                <td>
                    Storage:
                </td>
                <td>
                    <input type="radio" name="storage" value="0"
                    <%= form.storage == "0" ? " checked" : "" %>
                    >256GB
                    <input type="radio" name="storage" value="1"
                    <%= form.storage == "1" ? " checked" : "" %>
                    >512GB
                    <input type="radio" name="storage" value="2"
                    <%= form.storage == "2" ? " checked" : "" %>
                    >1TB
                </td>
            </tr>
            <tr>
                <td>Shipping:</td>
                <td>
                    <input type="checkbox" name="shipping" <%=
form.shipping ? " checked" : "" %>>
                </td>
            </tr>
            <tr>
                <td colspan="2">
                    <button type="submit">Buy</button>
                </td>
            </tr>
        </table>
    </form>
    <% if (price) { %>
        <p><span><%= model %></span> x <%= form.units %> purchased, Total
price is <b><%= price %></b></p>
    <% } %>
```

```
  <% if (Object.keys(errors).length > 0) { %>
        <div class="error">
            <h3>Errors</h3>
            <%- Object.values(errors).map(e => `<p>${e.msg}</p>`).join("")
%>
        </div>
    <% } %>
</body>
</html>
```

**Question 1 (60Pts):** Write a Web application file "***q1.php***" that shows the first three products initially. Each product shows the product image (under images folder) , name and the price. If there is a discount like 0.3, it means there is 30% off. In this case, show the original price with a line-through, and the new price underneath it. If there is no discount, just show the original price.

By clicking "prev" and "next" icons (***left.png*** and ***right.png***), the user can shift the product list by one product in left or right direction, respectively. The product list (**$products**) is given in ***db.php***.

The user clicks on a product name to select one. Its background color is marked with #CCF color. The user can select only one product at a time. Initially, none of the product is selected.

Do not forget to watch "***q1.mp4***" that demonstrates what you are supposed to do. You are NOT allowed to use Javascript. Hint: You can use a table with one row and five columns.

**Fig 1:** Initial page

**GET EXAMPLE 2:**

**q1.js:**

```
import express from "express";
import { products } from "./db.js";

const app = express();
// Serve static files from the "public" directory and set EJS as the view
engine
app.use(express.static("public"));
app.set("view engine", "ejs");

app.get("/", (req, res) => {
  // Get the 'pos' query parameter, default to 0 if not provided
  let pos = req.query.pos ?? 0
  pos = parseInt(pos)
  const { id } =  req.query

  // Flags to determine if "Previous" and "Next" buttons should be enabled
  const prevFlag = pos > 0
  const nextFlag = pos + 2 < products.length

  // Pagination logic: Get 3 products starting from the position 'pos'
  const paginatedProducts = products.slice(pos, pos + 3)

  res.render("q1", { products: paginatedProducts, pos, prevFlag, nextFlag,
id });
});

app.listen(3000, () => {
  console.log("Server is running on port 3000");
});
```

**q1.ejs:**

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        .red {
            color: red;
        }
        .grayout {
            color: gray;
            text-decoration: line-through;
        }
        section {
            display: flex;
            justify-content: space-between;
            gap: 20px;
        }
        section > div {
            width: 200px;
            border: 1px solid #CCC;
            border-radius: 5px;
            padding: 20px;
            text-align: center;
        }
        .left, .right {
            width: 100px;
            border: none ;
        }
        .price { font-size: 30px;;}
        .disabled {
            opacity: 0.2;
            pointer-events: none;
        }
        .selected {
            background-color: #FDD;
        }
    </style>
```

```
</head>
<body>
    <section>
        <div class="left">
            <% if (prevFlag) { %>
                <a href="/?pos=<%= pos - 1 %><%- id ? `&id=${id}` : ''
%>">
                    <img src="/images/left.png" >
                </a>
            <% } else { %>
                <img src="/images/left.png" class="disabled">
            <% } %>
        </div>
        <%-
           products.map(product => {
               return `
                   <div class="${id == product.id ? 'selected' : ''}">
                       <img src="/images/${product.image}" >
                       <h2><a
href="/?id=${product.id}&pos=${pos}">${product.name}</a></h2>
                       <p class="price">
                         ${
                           product.discount > 0 ?
                           `<span
class="grayout">${product.price}</span><br>
                            <span class="red">${product.price * (1 -
product.discount)}₺</span>`
                            : `<span class="red">${product.price}₺</span>`
                         }
                       </p>
                   </div>
               `
           }).join('')
        %>
        <div class="right">
            <% if (nextFlag) { %>
                <a href="/?pos=<%= pos + 1 %><%- id ? `&id=${id}` : ''
%>">
                    <img src="/images/right.png" >
                </a>
```

```
            <% } else { %>
                <img src="/images/right.png" class="disabled">
            <% } %>
        </div>

    </section>
</body>
</html>
```

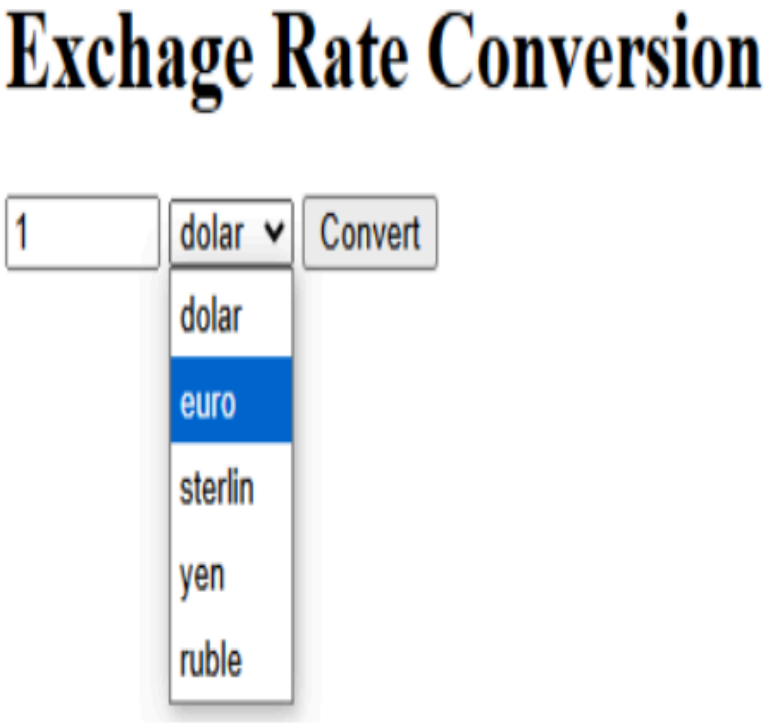


**Fig 2** : Web Form for the conversion

**POST EXAMPLE 2:**

q2.js

```
import express from "express";
import { body, validationResult } from "express-validator";
import { xrates } from "./db.js";
const app = express();
app.set("view engine", "ejs");

app.use(express.urlencoded({ extended: true }));

app.get("/", (req, res) => {
    res.render("q2", { xrates, form: {}, value: null, errorMsg: null });
})

app.post("/",
    body("amount").isFloat({ min: 0 }).withMessage("Error: The amount
should be a positive number"),
    (req, res) => {
        const errors = validationResult(req);
        let value = null, errorMsg = null;
        if (errors.isEmpty()) {
            const rate = xrates[req.body.currency];
            value = req.body.amount * rate.price;
        } else {
            errorMsg = errors.mapped().amount.msg;
        }
        res.render("q2", { xrates, form: req.body, value, errorMsg });
    }
);

app.listen(3000, () => console.log("Started on 3000"))
```

**q2.ejs:**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Form</title>
    <style>
        .error { color: red ; font-weight: bold;}
        .success { font-size: 2em; color: blue ;}
    </style>
</head>
<body>
    <form action="/" method="post">
        <h1>Exchage Rate Conversion</h1>
        <input type="text" name="amount" placeholder="Amount" size="5"
         value="<%= form.amount ?? "" %>"
        >
        <select name="currency">
            <% for (const rate of Object.entries(xrates)) { %>
                <option value="<%= rate[0] %>"
                  <%= form.currency == rate[0] ? " selected" : "" %>
                ><%= rate[1]["name"] %></option>
            <% } %>
        </select>
        <button>Convert</button>
    </form>
    <% if (value !== null) { %>
        <p class="success"><%= value %> TL</p>
    <% } %>

    <% if (errorMsg) { %>
        <p class="error"><%= errorMsg %></p>
    <% } %>
</body>
</html>
```